

Student:	Email:
Mohammed Anwar	anwarm@ferris.edu

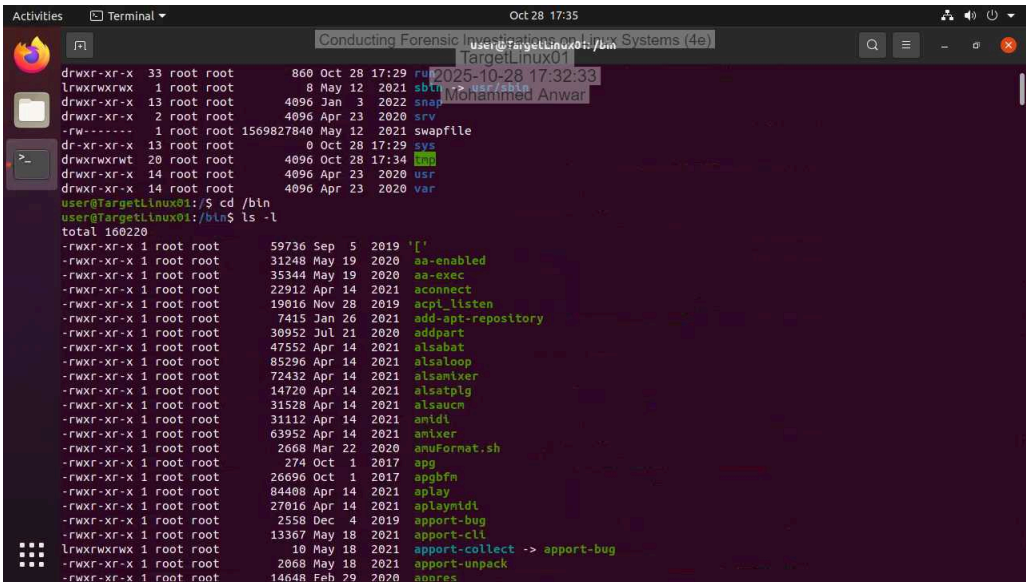
Time on Task:	Progress:
1 hour, 50 minutes	94%

Report Generated: Tuesday, October 28, 2025 at 6:23 PM

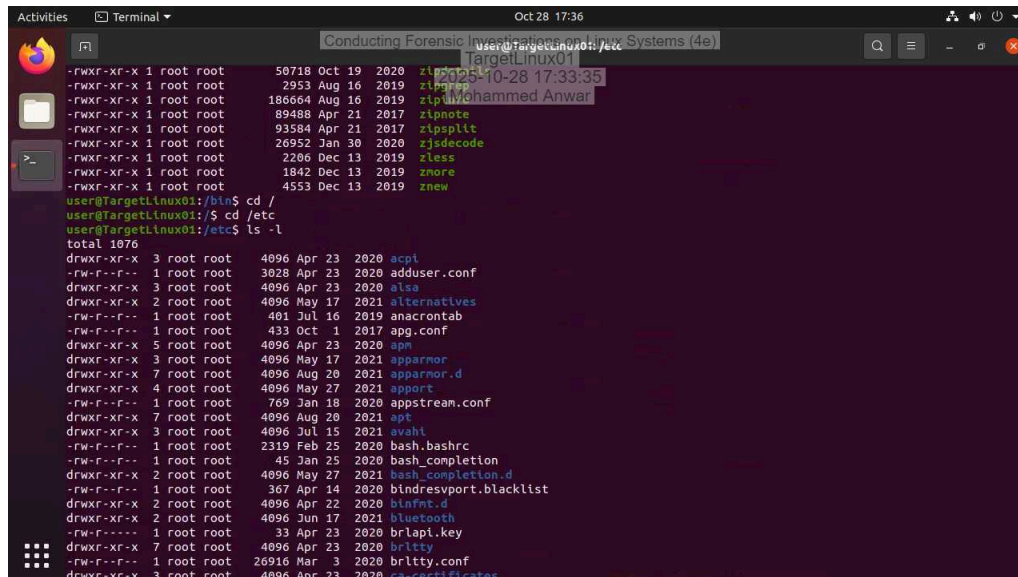
Section 1: Hands-On Demonstration

Part 1: Explore a Live Linux System

- 17. Make a screen capture showing the contents of the /bin directory.



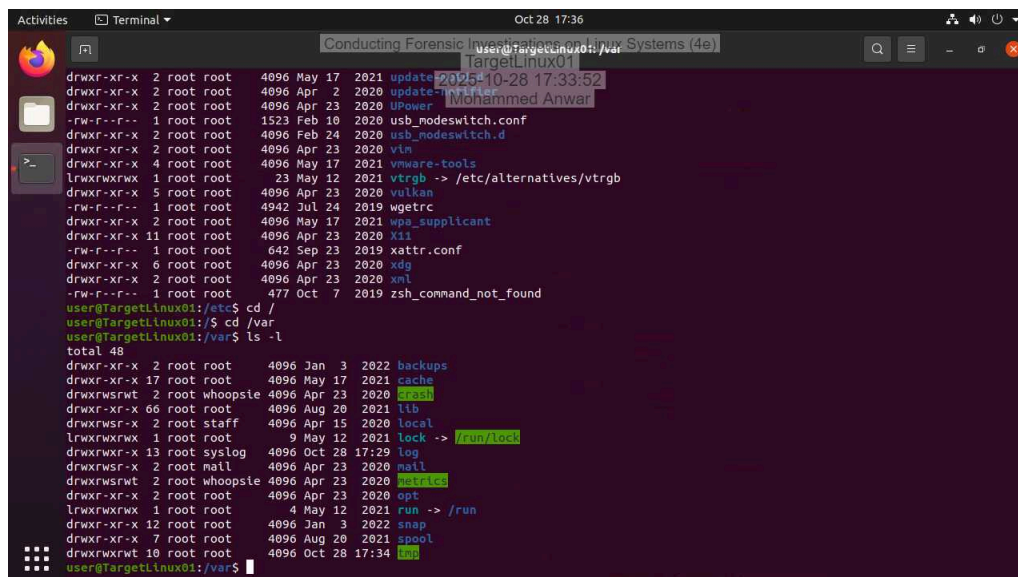
20. Make a screen capture showing the contents of the /etc directory.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the contents of the /etc directory. The user is logged in as "user@TargetLinux01". The terminal output shows the results of the command "ls -l /etc", displaying a list of files and directories with their permissions, owner, group, size, and modification date. The files listed include acpi, adduser.conf, alsa, alternatives, anacrontab, app.conf, apparmor, apparmor.d, appport, appstream.conf, apt, avahi, bash.bashrc, bash_completion, bash_completion.d, bindresolv.conf, blkid, bluetooth, brltty, brltty.conf, and ca-certificates.

```
user@TargetLinux01:~$ cd /
user@TargetLinux01:/$ cd /etc
user@TargetLinux01:/etc$ ls -l
total 1076
drwxr-xr-x 3 root root 4096 Apr 23 2020 acpi
-rw-r--r-- 1 root root 3028 Apr 23 2020 adduser.conf
drwxr-xr-x 3 root root 4096 Apr 23 2020 alsa
drwxr-xr-x 2 root root 4096 May 17 2021 alternatives
-rw-r--r-- 1 root root 401 Jul 16 2019 anacrontab
-rw-r--r-- 1 root root 433 Oct 1 2017 app.conf
drwxr-xr-x 5 root root 4096 Apr 23 2020 apparmor
drwxr-xr-x 3 root root 4096 May 17 2021 apparmor.d
drwxr-xr-x 7 root root 4096 Aug 20 2021 appport
drwxr-xr-x 4 root root 4096 May 27 2021 appstream.conf
-rw-r--r-- 1 root root 769 Jan 18 2020 appstream.conf
drwxr-xr-x 7 root root 4096 Aug 20 2021 apt
drwxr-xr-x 3 root root 4096 Jul 15 2021 avahi
-rw-r--r-- 1 root root 2319 Feb 25 2020 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 25 2020 bash_completion
drwxr-xr-x 2 root root 4096 May 27 2021 bash_completion.d
-rw-r--r-- 1 root root 367 Apr 14 2020 bindresolv.conf.blacklist
drwxr-xr-x 2 root root 4096 Apr 22 2020 blkid.d
drwxr-xr-x 2 root root 4096 Jun 17 2021 bluetooth
-rw-r----- 1 root root 33 Apr 23 2020 brltty.key
drwxr-xr-x 7 root root 4096 Apr 23 2020 brltty
-rw-r--r-- 1 root root 26916 Mar 3 2020 brltty.conf
drwxr-xr-x 3 root root 4096 Apr 23 2020 ca-certificates
```

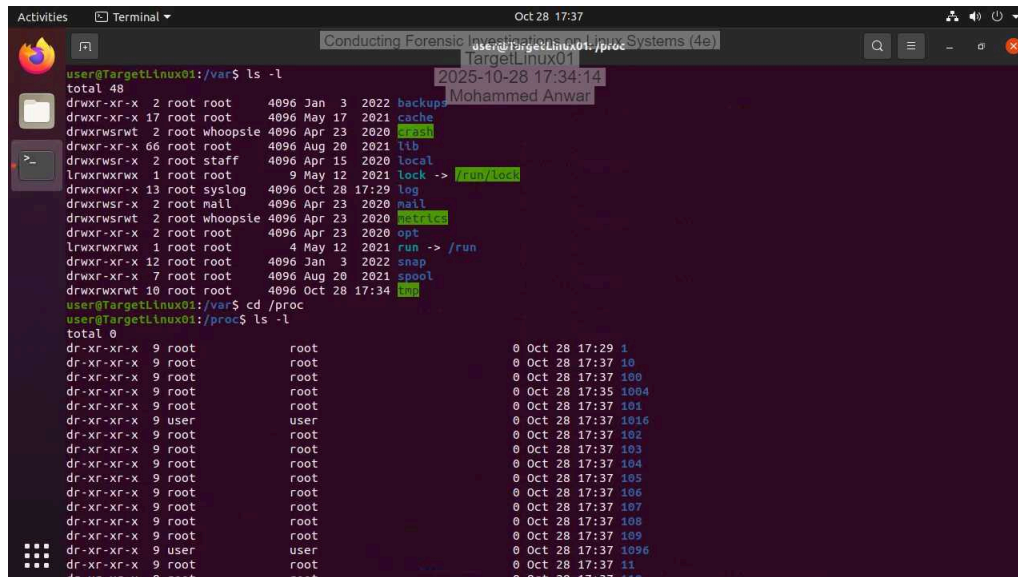
21. Make a screen capture showing the contents of the /var directory.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the contents of the /var directory. The user is logged in as "user@TargetLinux01". The terminal output shows the results of the command "ls -l /var", displaying a list of files and directories with their permissions, owner, group, size, and modification date. The files listed include backups, cache, cron, lib, local, lock, log, mail, mtrick, opt, run, snap, spool, and tmp.

```
user@TargetLinux01:/etc$ cd /
user@TargetLinux01:/$ cd /var
user@TargetLinux01:/var$ ls -l
total 48
drwxr-xr-x 2 root root 4096 Jan 3 2022 backups
drwxr-xr-x 17 root root 4096 May 17 2021 cache
drwxrwsrwt 2 root whoopste 4096 Apr 23 2020 cron
drwxr-xr-x 66 root root 4096 Aug 20 2021 lib
drwxrwsr-x 2 root staff 4096 Apr 15 2020 local
lrwxrwxrwx 1 root root 9 May 12 2021 lock -> /run/lock
drwxrwsr-x 13 root syslog 4096 Oct 28 17:29 log
drwxrwsr-x 2 root mail 4096 Apr 23 2020 mail
drwxrwsrwt 2 root whoopste 4096 Apr 23 2020 mtrick
drwxr-xr-x 2 root root 4096 Apr 23 2020 opt
lrwxrwxrwx 1 root root 4 May 12 2021 run -> /run
drwxr-xr-x 12 root root 4096 Jan 3 2022 snap
drwxr-xr-x 7 root root 4096 Aug 20 2021 spool
drwxrwsrwt 10 root root 4096 Oct 28 17:34 tmp
user@TargetLinux01:/var$
```

22. Make a screen capture showing the contents of the /proc directory.

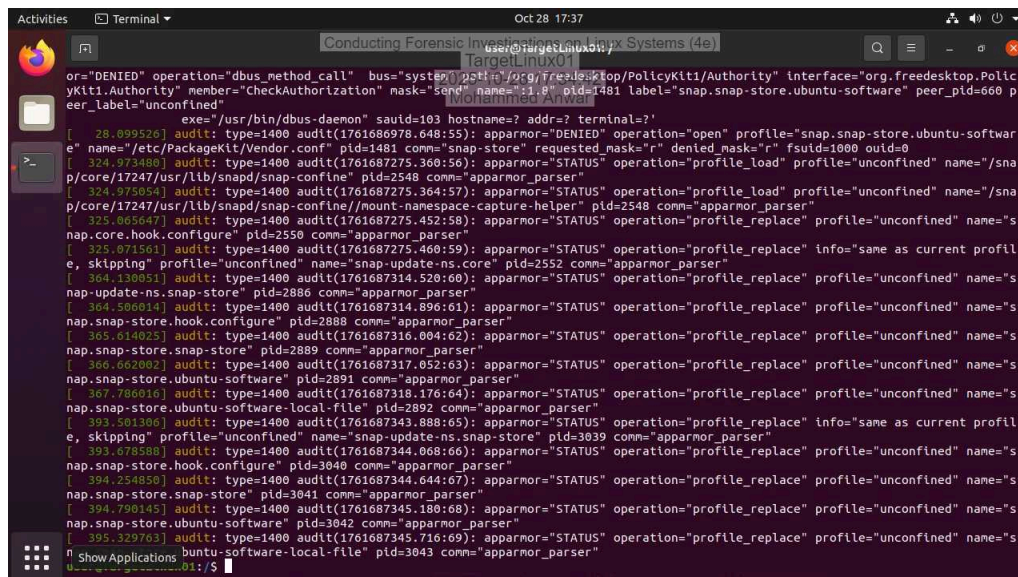


```
user@TargetLinux01:~$ ls -l /var
total 40
drwxr-xr-x 2 root root 4096 Jan 3 2022 backups
drwxr-xr-x 17 root root 4096 May 17 2021 cache
drwxrwsrwt 2 root whoopste 4096 Apr 23 2020 ceph
drwxr-xr-x 66 root root 4096 Aug 20 2021 lib
drwxrwsr-x 2 root staff 4096 Apr 15 2020 local
lrwxrwxrwx 1 root root 9 May 12 2021 lock -> /run/lock
drwxrwxr-x 13 root syslog 4096 Oct 28 17:29 log
drwxrwsr-x 2 root mail 4096 Apr 23 2020 mail
drwxrwsrwt 2 root whoopste 4096 Apr 23 2020 mnt
drwxr-xr-x 2 root root 4096 Apr 23 2020 opt
lrwxrwxrwx 1 root root 4 May 12 2021 run -> /run
drwxr-xr-x 12 root root 4096 Jan 3 2022 snap
drwxr-xr-x 7 root root 4096 Aug 20 2021 spool
drwxrwsrwt 10 root root 4096 Oct 28 17:34 tmp

user@TargetLinux01:~$ cd /proc
user@TargetLinux01:~$ ls -l /proc
total 0
dr-xr-xr-x 9 root root 0 Oct 28 17:29 1
dr-xr-xr-x 9 root root 0 Oct 28 17:37 10
dr-xr-xr-x 9 root root 0 Oct 28 17:37 100
dr-xr-xr-x 9 root root 0 Oct 28 17:35 1004
dr-xr-xr-x 9 root root 0 Oct 28 17:37 101
dr-xr-xr-x 9 user user 0 Oct 28 17:37 1016
dr-xr-xr-x 9 root root 0 Oct 28 17:37 102
dr-xr-xr-x 9 root root 0 Oct 28 17:37 103
dr-xr-xr-x 9 root root 0 Oct 28 17:37 104
dr-xr-xr-x 9 root root 0 Oct 28 17:37 105
dr-xr-xr-x 9 root root 0 Oct 28 17:37 106
dr-xr-xr-x 9 root root 0 Oct 28 17:37 107
dr-xr-xr-x 9 root root 0 Oct 28 17:37 108
dr-xr-xr-x 9 root root 0 Oct 28 17:37 109
dr-xr-xr-x 9 user user 0 Oct 28 17:37 1096
dr-xr-xr-x 9 root root 0 Oct 28 17:37 11
dr-xr-xr-x 9 root root 0 Oct 28 17:37 110
```

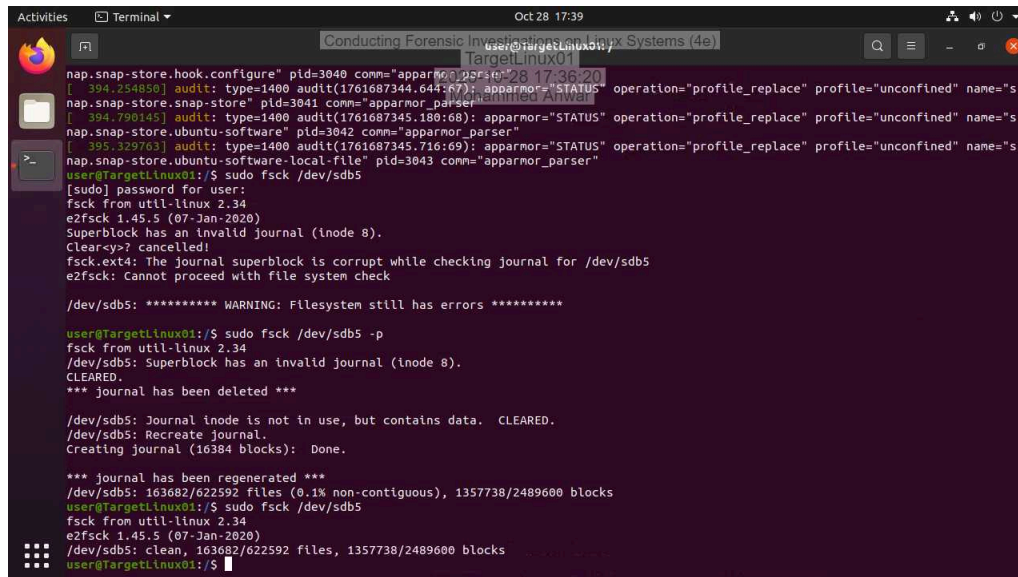
Part 2: Use Linux Shell Commands for Forensic Investigations

2. Make a screen capture showing the results of the dmesg command.



```
or="DENIED" operation="dbus_method_call" bus="system" path="/org/freedesktop/PolicyKit1/Authority" interface="org.freedesktop.PolicyKit1.Authority" member="CheckAuthorization" mask="send" name="s1.0" pid=1481 label="snap.snap-store.ubuntu-software" peer_pid=666 peer_label="unconfined"
exe="/usr/bin/dbus-daemon" sauid=103 hostname=? addr=? terminal=?
[ 28.899526] audit: type=1400 audit(1761686978.648:55): apparmor="DENIED" operation="open" profile="snap.snap-store.ubuntu-software" names="/etc/PackageKit/vendor.conf" pid=1481 comm="snap-store" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[ 324.973480] audit: type=1400 audit(1761687275.360:56): apparmor="STATUS" operation="profile_load" profile="unconfined" name="/snapd/core/17247/usr/lib/snapd/snap-confine" pid=2548 comm="apparmor_parser"
[ 324.975054] audit: type=1400 audit(1761687275.364:57): apparmor="STATUS" operation="profile_load" profile="unconfined" name="/snapd/core/17247/usr/lib/snapd/snap-confine/mount-namespace-capture-helper" pid=2548 comm="apparmor_parser"
[ 325.065647] audit: type=1400 audit(1761687275.452:58): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.core.hook.configure" pid=2550 comm="apparmor_parser"
[ 325.071561] audit: type=1400 audit(1761687275.460:59): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap-update-ns.core" pid=2552 comm="apparmor_parser"
[ 364.130051] audit: type=1400 audit(1761687314.520:60): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap-update-ns.snap-store" pid=2886 comm="apparmor_parser"
[ 364.500014] audit: type=1400 audit(1761687314.896:61): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=2888 comm="apparmor_parser"
[ 365.614025] audit: type=1400 audit(1761687316.004:62): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=2889 comm="apparmor_parser"
[ 366.662002] audit: type=1400 audit(1761687317.052:63): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=2891 comm="apparmor_parser"
[ 367.786016] audit: type=1400 audit(1761687318.176:64): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=2892 comm="apparmor_parser"
[ 393.601306] audit: type=1400 audit(1761687343.888:65): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap-update-ns.snap-store" pid=3039 comm="apparmor_parser"
[ 393.678588] audit: type=1400 audit(1761687344.068:66): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=3040 comm="apparmor_parser"
[ 394.254850] audit: type=1400 audit(1761687344.644:67): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=3041 comm="apparmor_parser"
[ 394.798145] audit: type=1400 audit(1761687345.180:68): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=3042 comm="apparmor_parser"
[ 395.329763] audit: type=1400 audit(1761687345.716:69): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=3043 comm="apparmor_parser"
user@TargetLinux01:~$
```

7. Make a screen capture showing the results of the fsck command.

A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the execution of the fsck command on the /dev/sdb5 partition. The output indicates a corrupted journal and a successful file system check after recreating the journal. The user is identified as "user@TargetLinux01".

```
user@TargetLinux01:~$ sudo fsck /dev/sdb5
[sudo] password for user:
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
Superblock has an invalid journal (inode 0).
Clearcys? cancelled!
fsck.ext4: The journal superblock is corrupt while checking journal for /dev/sdb5
e2fsck: Cannot proceed with file system check

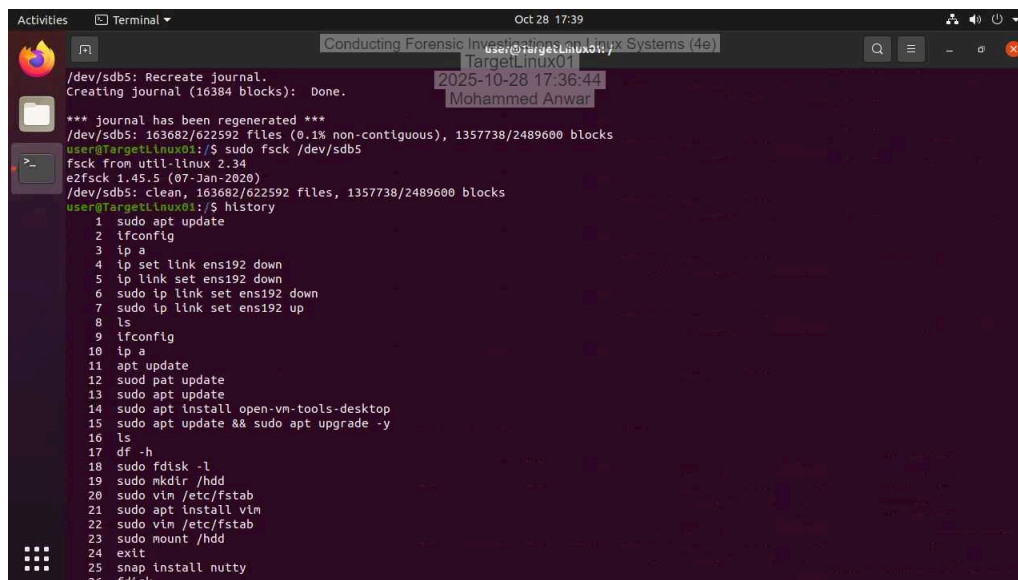
/dev/sdb5: ***** WARNING: Filesystem still has errors *****

user@TargetLinux01:~$ sudo fsck /dev/sdb5 -p
fsck from util-linux 2.34
/dev/sdb5: Superblock has an invalid journal (inode 0).
CLEARED.
*** journal has been deleted ***

/dev/sdb5: Journal inode is not in use, but contains data.  CLEARED.
/dev/sdb5: Recreate journal.
Creating journal (16384 blocks): Done.

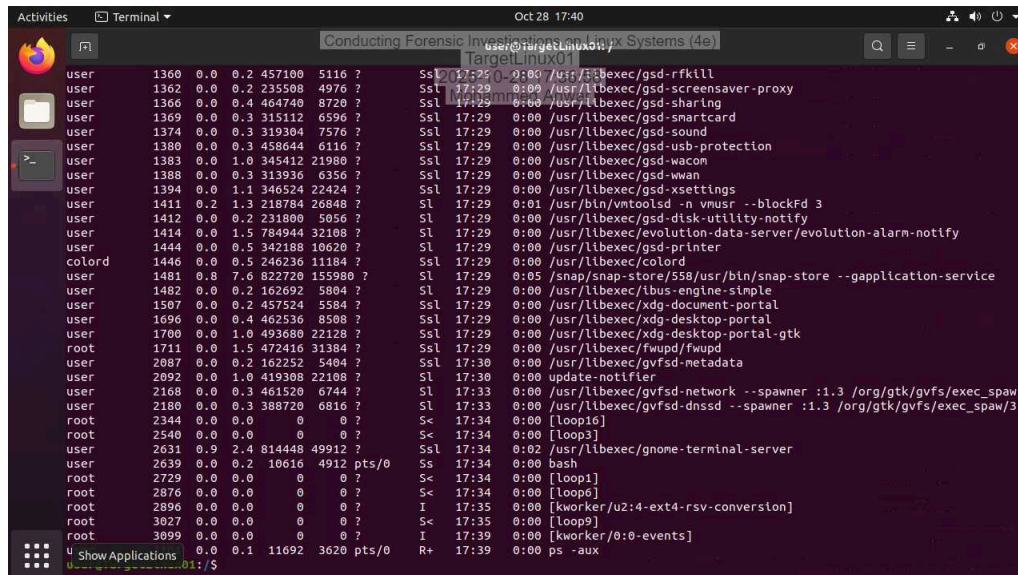
*** journal has been regenerated ***
/dev/sdb5: 163682/622592 files (0.1% non-contiguous), 1357738/2489600 blocks
user@TargetLinux01:~$ sudo fsck /dev/sdb5
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
/dev/sdb5: clean, 163682/622592 files, 1357738/2489600 blocks
user@TargetLinux01:~$
```

9. Make a screen capture showing the results of the history command.

A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the output of the history command. The list of commands includes system updates, network configuration, disk management, and file system checks. The user is identified as "user@TargetLinux01".

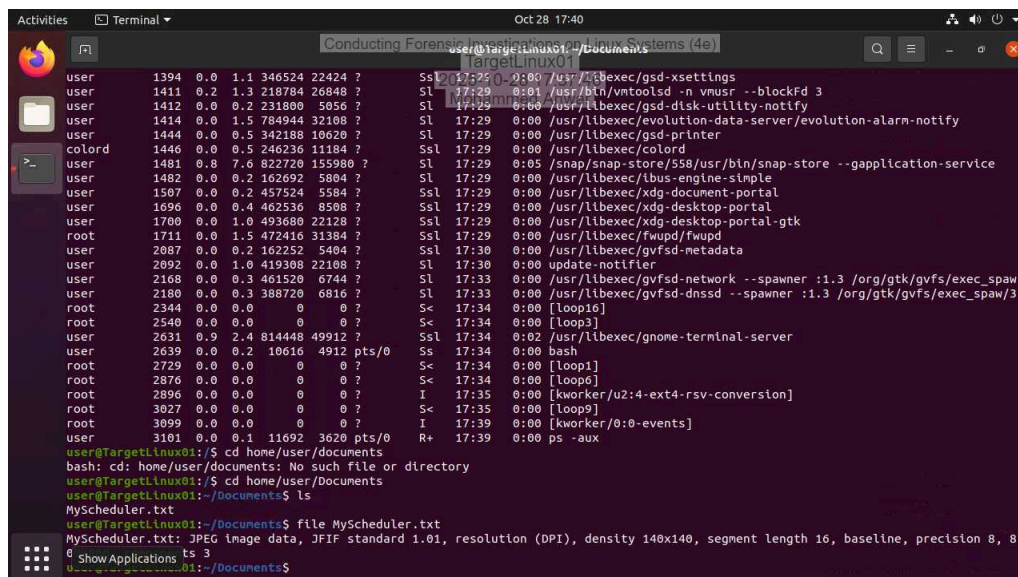
```
user@TargetLinux01:~$ history
1  sudo apt update
2  ifconfig
3  ip a
4  ip set link ens192 down
5  ip link set ens192 down
6  sudo ip link set ens192 down
7  sudo ip link set ens192 up
8  ls
9  ifconfig
10 ip a
11 apt update
12 suod pat update
13 sudo apt update
14 sudo apt install open-vm-tools-desktop
15 sudo apt update && sudo apt upgrade -y
16 ls
17 df -h
18 sudo fdisk -l
19 sudo mkdir /hdd
20 sudo vim /etc/fstab
21 sudo apt install vim
22 sudo vim /etc/fstab
23 sudo mount /hdd
24 exit
25 snap install nutty
26 fdisk
```


11. Make a screen capture showing the running processes.

A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the output of the 'ps -aux' command. The output lists various system processes including /usr/libexec/gsd-rfkill, /usr/libexec/gsd-screensaver-proxy, /usr/libexec/gsd-sharing, /usr/libexec/gsd-smartcard, /usr/libexec/gsd-sound, /usr/libexec/gsd-usb-protection, /usr/libexec/gsd-wacom, /usr/libexec/gsd-wwan, /usr/libexec/gsd-xsettings, /usr/bin/vmtoolsd -n vmusr --blockFd 3, /usr/libexec/gsd-disk-utility-notify, /usr/libexec/evolution-data-server/evolution-alarm-notify, /usr/libexec/gsd-printer, /usr/libexec/colord, /snap/snap-store/558/usr/bin/snap-store --gapplication-service, /usr/libexec/lbus-engine-simple, /usr/libexec/xdg-document-portal, /usr/libexec/xdg-desktop-portal, /usr/libexec/xdg-desktop-portal-gtk, /usr/libexec/fwupd/fwupd, /usr/libexec/gvfsd-metadata, update-notifier, /usr/libexec/gvfsd-network --spawner :1.3 /org/gtk/gvfs/exec_spaw, /usr/libexec/gvfsd-dnssd --spawner :1.3 /org/gtk/gvfs/exec_spaw/3, [loop16], [loop3], /usr/libexec/gnome-terminal-server, bash, [loop1], [loop6], [kworker/u2:4-ext4-rsv-conversion], [loop9], [kworker/0:0-events], and ps -aux.

```
user 1360 0.0 0.2 457100 5116 ? Ssl 17:29 0:00 /usr/libexec/gsd-rfkill
user 1362 0.0 0.2 235908 4976 ? Ssl 17:29 0:00 /usr/libexec/gsd-screensaver-proxy
user 1366 0.0 0.4 404740 8720 ? Ssl 17:29 0:00 /usr/libexec/gsd-sharing
user 1369 0.0 0.3 315112 6596 ? Ssl 17:29 0:00 /usr/libexec/gsd-smartcard
user 1374 0.0 0.3 319304 7576 ? Ssl 17:29 0:00 /usr/libexec/gsd-sound
user 1380 0.0 0.3 458644 6116 ? Ssl 17:29 0:00 /usr/libexec/gsd-usb-protection
user 1383 0.0 1.0 345412 21980 ? Ssl 17:29 0:00 /usr/libexec/gsd-wacom
user 1388 0.0 0.3 313936 6356 ? Ssl 17:29 0:00 /usr/libexec/gsd-wwan
user 1394 0.0 1.1 346524 22424 ? Ssl 17:29 0:00 /usr/libexec/gsd-xsettings
user 1411 0.2 1.3 218784 26848 ? Sl 17:29 0:01 /usr/bin/vmtoolsd -n vmusr --blockFd 3
user 1412 0.0 0.2 231000 5056 ? Sl 17:29 0:00 /usr/libexec/gsd-disk-utility-notify
user 1414 0.0 1.5 784944 32100 ? Sl 17:29 0:00 /usr/libexec/evolution-data-server/evolution-alarm-notify
user 1444 0.0 0.5 342188 10620 ? Sl 17:29 0:00 /usr/libexec/gsd-printer
colord 1446 0.0 0.5 246236 11184 ? Ssl 17:29 0:00 /usr/libexec/colord
user 1481 0.8 7.6 822720 155980 ? Sl 17:29 0:05 /snap/snap-store/558/usr/bin/snap-store --gapplication-service
user 1482 0.0 0.2 162692 5804 ? Sl 17:29 0:00 /usr/libexec/lbus-engine-simple
user 1507 0.0 0.2 457524 5584 ? Ssl 17:29 0:00 /usr/libexec/xdg-document-portal
user 1696 0.0 0.4 462536 8508 ? Ssl 17:29 0:00 /usr/libexec/xdg-desktop-portal
user 1700 0.0 1.0 493680 22128 ? Ssl 17:29 0:00 /usr/libexec/xdg-desktop-portal-gtk
root 1711 0.0 1.5 472416 31384 ? Ssl 17:29 0:00 /usr/libexec/fwupd/fwupd
user 2087 0.0 0.2 162252 5404 ? Ssl 17:30 0:00 /usr/libexec/gvfsd-metadata
user 2092 0.0 1.0 419308 22100 ? Sl 17:30 0:00 update-notifier
user 2168 0.0 0.3 461520 6744 ? Sl 17:33 0:00 /usr/libexec/gvfsd-network --spawner :1.3 /org/gtk/gvfs/exec_spaw
user 2180 0.0 0.3 388720 6816 ? Sl 17:33 0:00 /usr/libexec/gvfsd-dnssd --spawner :1.3 /org/gtk/gvfs/exec_spaw/3
root 2344 0.0 0.0 0 0 ? S< 17:34 0:00 [loop16]
root 2540 0.0 0.0 0 0 ? S< 17:34 0:00 [loop3]
user 2631 0.9 2.4 814448 49912 ? Ssl 17:34 0:02 /usr/libexec/gnome-terminal-server
user 2639 0.0 0.2 106616 4912 pts/0 Ss 17:34 0:00 bash
root 2729 0.0 0.0 0 0 ? S< 17:34 0:00 [loop1]
root 2876 0.0 0.0 0 0 ? S< 17:34 0:00 [loop6]
root 2896 0.0 0.0 0 0 ? I 17:35 0:00 [kworker/u2:4-ext4-rsv-conversion]
root 3027 0.0 0.0 0 0 ? S< 17:35 0:00 [loop9]
root 3099 0.0 0.0 0 0 ? I 17:39 0:00 [kworker/0:0-events]
user 3101 0.0 0.1 11692 3620 pts/0 R+ 17:39 0:00 ps -aux
```

15. Make a screen capture showing the results of the file command.

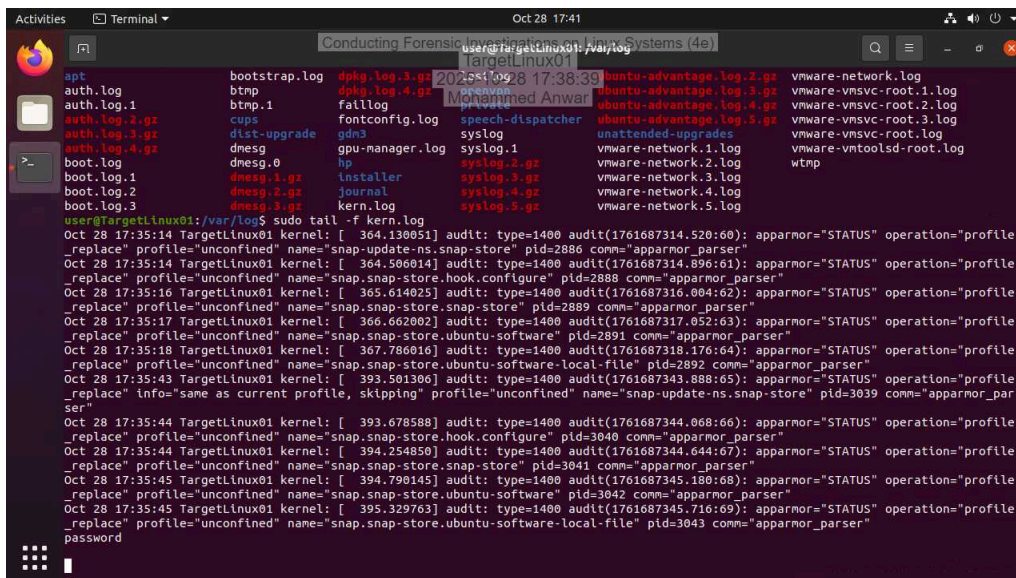
A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the output of the 'ls' command in the /Documents directory. The output lists files: MyScheduler.txt, MyScheduler.txt, and 3. The terminal also shows the command 'file MyScheduler.txt' and its output: 'JPEG image data, JFIF standard 1.01, resolution (DPI), density 140x140, segment length 16, baseline, precision 8, 8'.

```
user 1394 0.0 1.1 346524 22424 ? Ssl 17:29 0:00 /usr/libexec/gsd-xsettings
user 1411 0.2 1.3 218784 26848 ? Sl 17:29 0:01 /usr/bin/vmtoolsd -n vmusr --blockFd 3
user 1412 0.0 0.2 231000 5056 ? Sl 17:29 0:00 /usr/libexec/gsd-disk-utility-notify
user 1414 0.0 1.5 784944 32100 ? Sl 17:29 0:00 /usr/libexec/evolution-data-server/evolution-alarm-notify
user 1444 0.0 0.5 342188 10620 ? Sl 17:29 0:00 /usr/libexec/gsd-printer
colord 1446 0.0 0.5 246236 11184 ? Ssl 17:29 0:00 /usr/libexec/colord
user 1481 0.8 7.6 822720 155980 ? Sl 17:29 0:05 /snap/snap-store/558/usr/bin/snap-store --gapplication-service
user 1482 0.0 0.2 162692 5804 ? Sl 17:29 0:00 /usr/libexec/lbus-engine-simple
user 1507 0.0 0.2 457524 5584 ? Ssl 17:29 0:00 /usr/libexec/xdg-document-portal
user 1696 0.0 0.4 462536 8508 ? Ssl 17:29 0:00 /usr/libexec/xdg-desktop-portal
user 1700 0.0 1.0 493680 22128 ? Ssl 17:29 0:00 /usr/libexec/xdg-desktop-portal-gtk
root 1711 0.0 1.5 472416 31384 ? Ssl 17:29 0:00 /usr/libexec/fwupd/fwupd
user 2087 0.0 0.2 162252 5404 ? Ssl 17:30 0:00 /usr/libexec/gvfsd-metadata
user 2092 0.0 1.0 419308 22100 ? Sl 17:30 0:00 update-notifier
user 2168 0.0 0.3 461520 6744 ? Sl 17:33 0:00 /usr/libexec/gvfsd-network --spawner :1.3 /org/gtk/gvfs/exec_spaw
user 2180 0.0 0.3 388720 6816 ? Sl 17:33 0:00 /usr/libexec/gvfsd-dnssd --spawner :1.3 /org/gtk/gvfs/exec_spaw/3
root 2344 0.0 0.0 0 0 ? S< 17:34 0:00 [loop16]
root 2540 0.0 0.0 0 0 ? S< 17:34 0:00 [loop3]
user 2631 0.9 2.4 814448 49912 ? Ssl 17:34 0:02 /usr/libexec/gnome-terminal-server
user 2639 0.0 0.2 106616 4912 pts/0 Ss 17:34 0:00 bash
root 2729 0.0 0.0 0 0 ? S< 17:34 0:00 [loop1]
root 2876 0.0 0.0 0 0 ? S< 17:34 0:00 [loop6]
root 2896 0.0 0.0 0 0 ? I 17:35 0:00 [kworker/u2:4-ext4-rsv-conversion]
root 3027 0.0 0.0 0 0 ? S< 17:35 0:00 [loop9]
root 3099 0.0 0.0 0 0 ? I 17:39 0:00 [kworker/0:0-events]
user 3101 0.0 0.1 11692 3620 pts/0 R+ 17:39 0:00 ps -aux

user@TargetLinux01:/$ cd home/user/documents
bash: cd: home/user/documents: No such file or directory
user@TargetLinux01:/$ cd home/user/Documents
user@TargetLinux01:~/Documents$ ls
MyScheduler.txt
user@TargetLinux01:~/Documents$ file MyScheduler.txt
MyScheduler.txt: JPEG image data, JFIF standard 1.01, resolution (DPI), density 140x140, segment length 16, baseline, precision 8, 8
3
user@TargetLinux01:~/Documents$
```

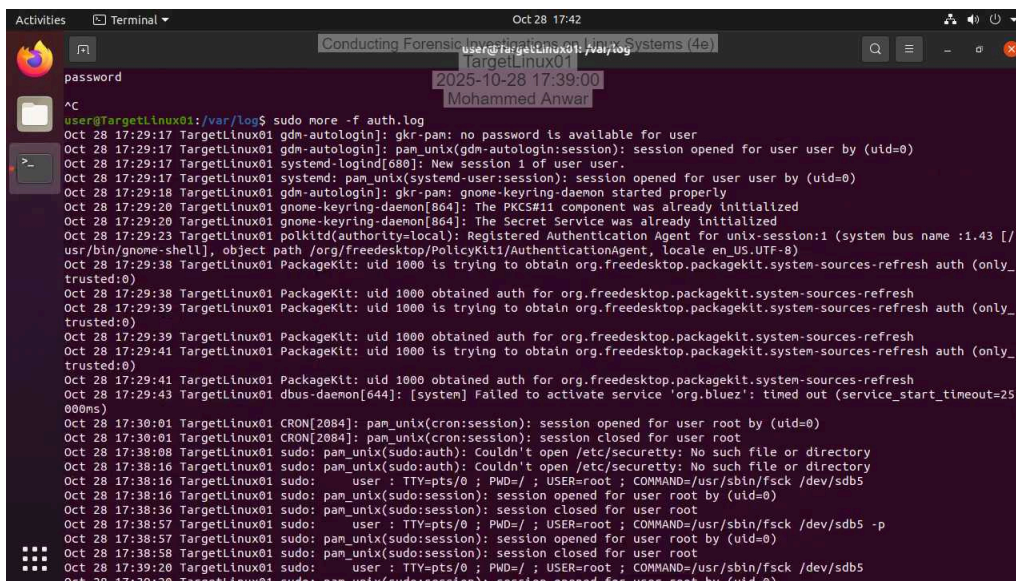
Part 3: Retrieve Logs Files on a Live Linux System

4. Make a screen capture showing the records in the kern.log file.



```
user@TargetLinux01: /var/log$ sudo tail -f kern.log
Oct 28 17:35:14 TargetLinux01 kernel: [ 364.130051] audit: type=1400 audit(1761687314.520:60): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.update-ns.snap-store" pid=2886 comm="apparmor_parser"
Oct 28 17:35:14 TargetLinux01 kernel: [ 364.506014] audit: type=1400 audit(1761687314.896:61): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=2888 comm="apparmor_parser"
Oct 28 17:35:16 TargetLinux01 kernel: [ 365.614025] audit: type=1400 audit(1761687316.004:62): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=2889 comm="apparmor_parser"
Oct 28 17:35:17 TargetLinux01 kernel: [ 366.602002] audit: type=1400 audit(1761687317.052:63): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=2891 comm="apparmor_parser"
Oct 28 17:35:18 TargetLinux01 kernel: [ 367.786016] audit: type=1400 audit(1761687318.176:64): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=2892 comm="apparmor_parser"
Oct 28 17:35:43 TargetLinux01 kernel: [ 393.501306] audit: type=1400 audit(1761687343.888:65): apparmor="STATUS" operation="profile
_replace" info="same as current profile, skipping" profile="unconfined" name="snap.update-ns.snap-store" pid=3039 comm="apparmor_par
ser"
Oct 28 17:35:44 TargetLinux01 kernel: [ 393.678588] audit: type=1400 audit(1761687344.068:66): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=3040 comm="apparmor_parser"
Oct 28 17:35:44 TargetLinux01 kernel: [ 394.254850] audit: type=1400 audit(1761687344.644:67): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=3041 comm="apparmor_parser"
Oct 28 17:35:45 TargetLinux01 kernel: [ 394.790145] audit: type=1400 audit(1761687345.180:68): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=3042 comm="apparmor_parser"
Oct 28 17:35:45 TargetLinux01 kernel: [ 395.329763] audit: type=1400 audit(1761687345.716:69): apparmor="STATUS" operation="profile
_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=3043 comm="apparmor_parser"
password
```

7. Make a screen capture showing the records in the auth.log file.



```
password
^C
user@TargetLinux01: /var/log$ sudo more -f auth.log
Oct 28 17:29:17 TargetLinux01 gdm-autologin]: gkr-pam: no password is available for user
Oct 28 17:29:17 TargetLinux01 gdm-autologin]: pam_unix(gdm-autologin:session): session opened for user user by (uid=0)
Oct 28 17:29:17 TargetLinux01 systemd-logind[680]: New session 1 of user user.
Oct 28 17:29:17 TargetLinux01 systemd: pam_unix(systemd-user:session): session opened for user user by (uid=0)
Oct 28 17:29:18 TargetLinux01 gdm-autologin]: gkr-pam: gnome-keyring-daemon started properly
Oct 28 17:29:20 TargetLinux01 gnome-keyring-daemon[864]: The PKCS#11 component was already initialized
Oct 28 17:29:20 TargetLinux01 gnome-keyring-daemon[864]: The Secret Service was already initialized
Oct 28 17:29:23 TargetLinux01 polkitd(authority=local): Registered Authentication Agent for unix-session:1 (system bus name :1.43 [/
usr/bin/gnome-shell], object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8)
Oct 28 17:29:38 TargetLinux01 PackageKit: uid 1000 is trying to obtain org.freedesktop.packagekit.system-sources-refresh auth (only_
trusted:0)
Oct 28 17:29:38 TargetLinux01 PackageKit: uid 1000 obtained auth for org.freedesktop.packagekit.system-sources-refresh
Oct 28 17:29:39 TargetLinux01 PackageKit: uid 1000 is trying to obtain org.freedesktop.packagekit.system-sources-refresh auth (only_
trusted:0)
Oct 28 17:29:39 TargetLinux01 PackageKit: uid 1000 obtained auth for org.freedesktop.packagekit.system-sources-refresh
Oct 28 17:29:41 TargetLinux01 PackageKit: uid 1000 is trying to obtain org.freedesktop.packagekit.system-sources-refresh auth (only_
trusted:0)
Oct 28 17:29:41 TargetLinux01 PackageKit: uid 1000 obtained auth for org.freedesktop.packagekit.system-sources-refresh
Oct 28 17:29:43 TargetLinux01 dbus-daemon[644]: [system] Failed to activate service 'org.bluez': timed out (service_start_timeout=25
000ms)
Oct 28 17:30:01 TargetLinux01 CRON[2084]: pam_unix(cron:session): session opened for user root by (uid=0)
Oct 28 17:30:01 TargetLinux01 CRON[2084]: pam_unix(cron:session): session closed for user root
Oct 28 17:38:08 TargetLinux01 sudo: pam_unix(sudo:auth): Couldn't open /etc/securetty: No such file or directory
Oct 28 17:38:16 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/ ; USER=root ; COMMAND=/usr/sbin/fsck /dev/sdb5
Oct 28 17:38:16 TargetLinux01 sudo: pam_unix(sudo:session): session opened for user root by (uid=0)
Oct 28 17:38:30 TargetLinux01 sudo: pam_unix(sudo:session): session closed for user root
Oct 28 17:38:57 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/ ; USER=root ; COMMAND=/usr/sbin/fsck /dev/sdb5 -p
Oct 28 17:38:57 TargetLinux01 sudo: pam_unix(sudo:session): session opened for user root by (uid=0)
Oct 28 17:38:58 TargetLinux01 sudo: pam_unix(sudo:session): session closed for user root
Oct 28 17:39:20 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/ ; USER=root ; COMMAND=/usr/sbin/fsck /dev/sdb5
Oct 28 17:39:20 TargetLinux01 sudo: pam_unix(sudo:session): session opened for user root by (uid=0)
```

Section 2: Applied Learning

Part 1: Identify Login Attempts on a Linux Drive Image

15. **Document** the names of the two non-root users that attempted to log in, the number of attempts detected, the date/time range of the attempts, the source IP address for the login attempts, and the port.

noel and pam

17. **Document** the date and time the most recent successful login for the user(s) that you previously identified in step 15.

both noel and pam failed to login

Part 2: Identify Software Installations on a Linux Drive Image

3. **Document** the applications that were installed using apt-get, then use the Internet to identify the ones that might be considered suspicious.

apt install logkeys

Part 3: Identify External Drive Attachments on a Linux Drive Image

4. **Document** when the USB storage device was connected and its serial number.

June 10th at 10:24:12

